

p0408r1 - Efficient Access to `basic_stringbuf`'s Buffer

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2017-02-03

Document Number:	p0408r1
Date:	2017-02-03
Project:	Programming Language C++
Audience:	LWG/LEWG
Target:	C++20

1 Motivation

Streams have been the oldest part of the C++ standard library and their specification doesn't take into account many things introduced since C++11. One of the oversights is that there is no non-copying access to the internal buffer of a `basic_stringbuf` which makes at least the obtaining of the output results from an `ostreamstream` inefficient, because a copy is always made. I personally speculate that this was also the reason why `basic_strbuf` took so long to get deprecated with its `char *` access.

With move semantics and `basic_string_view` there is no longer a reason to keep this pessimisation alive on `basic_stringbuf`.

[I also believe we should remove `basic\_strbuf` from the standard's appendix \[depr.str.strstreams\].](#)

2 Introduction

This paper proposes to adjust the API of `basic_stringbuf` and the corresponding stream class templates to allow accessing the underlying string more efficiently.

C++17 and library TS have `basic_string_view` allowing an efficient read-only access to a contiguous sequence of characters which I believe `basic_stringbuf` has to guarantee about its internal buffer, even if it is not implemented using `basic_string` obtaining a `basic_string_view` on the internal buffer should work sidestepping the copy overhead of calling `str()`.

On the other hand, there is no means to construct a `basic_string` and move from it into a `basic_stringbuf` via a constructor or a move-enabled overload of `str(basic_string &&)`.

2.1 History

Discussed in LEWG Issaquah. Answering some questions and raising more. Reflected in this paper.

2.1.1 Changes from r0

- Added more context to synopsis sections to see all overloads (Thanks Alisdair).
- rename `str_view()` to just `view()`. There was discussion on including an explicit conversion operator as well, but I didn't add it yet (my implementation has it).
- renamed r-value-ref qualified `str()` to `pilfer()` and removed the reference qualification from it and remaining `str()` member.
- Added allocator parameter for the `basic_string` parameter/result to member functions (see p0407 for allocator support for stringstream in general)

3 Acknowledgements

- Daniel Krüger encouraged me to pursue this track.
- [Alisdair Meredith for telling me to include context in the synopsis showing all overloads. That is the only change in this version, no semantic changes!](#)

4 Impact on the Standard

This is an extension to the API of `basic_stringbuf`, `basic_stringstream`, `basic_istringstream`, and `basic_ostringstream` class templates.

[This paper addresses both Library Fundamentals TS 3 and C++Next \(2020?\). When added to the standard draft section \[depr.str.strstreams\] should be removed from it.](#)

5 Design Decisions

After experimentation I decided that substituting the `(basic_string<charT, traits, Allocator const &)` constructors in favor of passing a `basic_string_view` would lead to ambiguities with the new move-from-string constructors.

5.1 Open Issues discussed by LEWG in Issquah

- Is the name of the `str_view()` member function ok? No. Renamed to `view()`
- Should the `str()&&` overload be provided for move-out? No. give it another name (`pilfer`) and remove rvalue-ref-qualification.
- Should `str()&&` empty the character sequence or leave it in an unspecified but valid state? Empty it, and specify.
- Provide guidance on validity lifetime of of the obtained `string_view` object.

6 Technical Specifications

The following is relative to n4604.

Remove section on `char*` streams [depr.str.strstreams] and all its subsections from appendix D.

6.1 27.8.2 Adjust synopsis of `basic_stringbuf` [`stringbuf`]

Add a new constructor overload:

```
// ??, constructors:
explicit basic_stringbuf(
    ios_base::openmode which = ios_base::in | ios_base::out);
template<class SAlloc=Allocator>
explicit basic_stringbuf(
    const basic_string<charT, traits, SAllocator>& str,
    ios_base::openmode which = ios_base::in | ios_base::out);

explicit basic_stringbuf(
    basic_string<charT, traits, Allocator>&& s,
    ios_base::openmode which = ios_base::in | ios_base::out);

basic_stringbuf(const basic_stringbuf& rhs) = delete;
basic_stringbuf(basic_stringbuf&& rhs);
```

Change the getting `str()` overload to take an `Allocator` for the returned string. Change the `str()` overload copying into the string buffer to take an allocator template parameter that could differ from the buffer's own `Allocator`. Add a `str()` overload that moves from its string rvalue-reference argument into the internal buffer. Add the `pilfer()` member function obtaining a string from the internal buffer by moving from it. Add the `view()` member function obtaining a `string_view` to the underlying internal buffer.

```
// ??, get and set:
template<class SAlloc=Allocator>
basic_string<charT,traits,SAllocator> str(const SAlloc& sa=SAlloc()) const;
template<class SAlloc=Allocator>
void str(const basic_string<charT, traits, SAllocator>& s);

void str(basic_string<charT, traits, Allocator>&& s);
basic_string<charT,traits,Allocator> pilfer();
basic_string_view<charT, traits> view() const;
```

6.1.1 27.8.2.1 basic_stringbuf constructors [stringbuf.cons]

Modify the following constructor specification:

```
template<class SAlloc=Allocator>
explicit basic_stringbuf(
    const basic_string<charT, traits, SAllocator>& str,
    ios_base::openmode which = ios_base::in | ios_base::out);
```

- 1 *Effects:* Constructs an object of class `basic_stringbuf`, initializing the base class with `basic_streambuf()` (??), and initializing mode with `which`. Then calls `str(s)`.

Add the following constructor specification:

```
explicit basic_stringbuf(
    basic_string<charT, traits, Allocator>&& s,
    ios_base::openmode which = ios_base::in | ios_base::out);
```

- 2 *Effects:* Constructs an object of class `basic_stringbuf`, initializing the base class with `basic_streambuf()` (27.6.3.1), and initializing mode with `which`. Then calls `str(std::move(s))`.

Note to editors: if p0407 is accepted the changes there for allocators apply here as well. However, different allocators for `s` and the `basic_stringbuf` will result in a copy instead of a move.

6.1.2 27.8.2.3 Member functions [stringbuf.members]

Add an allocator parameter for the copied from string to allow having a different allocator than the underlying stream.

```
template<class SAlloc=Allocator>
basic_string<charT,traits,SAllocator> str(const SAlloc& sa=SAlloc()) const;
```

Change p1 to use plural for "str(basic_string) member functions" and refer to the allocator:

- 1 *Returns:* A `basic_string` object with allocator `sa` whose content is equal to the `basic_stringbuf` underlying character sequence. If the `basic_stringbuf` was created only in input mode, the resultant `basic_string` contains the character sequence in the range `[eback(), egptr())`. If the `basic_stringbuf` was created with `which & ios_base::out` being true then the resultant `basic_string` contains the character sequence in the range `[pbase(), high_mark)`, where `high_mark` represents the position one past the highest initialized character in the buffer. Characters can be initialized by writing to the stream, by constructing the `basic_stringbuf` with a `basic_string`, or by calling one of the `str(basic_string)` member functions. In the case of calling one of the `str(basic_string)` member functions, all characters initialized prior to the call are now considered uninitialized (except for those characters re-initialized by the new `basic_string`). Otherwise the `basic_stringbuf` has been created in neither input nor output mode and a zero length `basic_string` is returned.

Add the following specifications and adjust the wording of `str() const` according to the wording given for `view() const` member function.:

```
void str(basic_string<charT, traits, Allocator>&& s);
```

- 2 *Effects:* Moves the content of `s` into the `basic_stringbuf` underlying character sequence and initializes the input and output sequences according to mode.
- 3 *Postconditions:* Let `size` denote the original value of `s.size()` before the move. If `mode & ios_base::out` is true, `pbase()` points to the first underlying character and `egptr() >= pbase() + size` holds; in addition, if `mode & ios_base::ate` is true, `pptr() == pbase() + size` holds, otherwise `pptr() == pbase()` is true. If `mode & ios_base::in` is true, `eback()` points to the first underlying character, and both `gptra() == eback()` and `egptr() == eback() + size` hold.

```
basic_string<charT, traits, Allocator> pilfer();
```

- 4 *Returns:* A `basic_string` object moved from the `basic_stringbuf` underlying character sequence. If the `basic_stringbuf` was created only in input mode, `basic_string(eback(), egptr()-eback())`. If the `basic_stringbuf` was created with `which & ios_base::out` being true then `basic_string(pbase(), high_mark-pbase())`, where `high_mark` represents the position one past the highest initialized character in the buffer. Characters can be initialized by writing to the stream, by constructing the `basic_stringbuf` with a `basic_string`, or by calling one of the `str(basic_string)` member functions. In the case of calling one of the `str(basic_string)` member functions, all characters initialized prior to the call are now considered uninitialized (except for those characters re-initialized by the new `basic_string`). Otherwise the `basic_stringbuf` has been created in neither input nor output mode and an empty `basic_string` is returned.
- 5 *Postcondition:* The underlying character sequence is empty.

```
basic_string_view<charT, traits> view() const;
```

- 6 *Returns:* A `basic_string_view` object referring to the `basic_stringbuf` underlying character sequence. If the `basic_stringbuf` was created only in input mode, `basic_string_view(eback(), egptr()-eback())`. If the `basic_stringbuf` was created with `which & ios_base::out` being true then `basic_string_view(pbase(), high_mark-pbase())`, where `high_mark` represents the position one past the highest initialized character in the buffer. Characters can be initialized by writing to the stream, by constructing the `basic_stringbuf` with a `basic_string`, or by calling one of the `str(basic_string)` member functions. In the case of calling one of the `str(basic_string)` member functions, all characters initialized prior to the call are now considered uninitialized (except for those characters re-initialized by the new `basic_string`). Otherwise the `basic_stringbuf` has been created in neither input nor output mode and a `basic_string_view` referring to an empty range is returned.
- 7 [Note: Using the returned `basic_string_view` object after destruction or any modification of the character sequence underlying `*this`, such as output on the holding stream, will cause undefined behavior, because the internal string referred by the return value might have changed or re-allocated. — end note]

6.2 27.8.3 Adjust synopsis of `basic_istream` [`istream`]

Add a new constructor overload and change the one taking the string by copy to allow a different allocator for the copied from string:

```
// ??, constructors:
explicit basic_istream(
    ios_base::openmode which = ios_base::in);
template<class SAlloc=Allocator>
explicit basic_istream(
    const basic_string<charT, traits, SAllocator>& str,
    ios_base::openmode which = ios_base::in);

explicit basic_istream(
    basic_string<charT, traits, Allocator>&& str,
    ios_base::openmode which = ios_base::in);

basic_istream(const basic_istream& rhs) = delete;
basic_istream(basic_istream&& rhs);
```

Change the `str()` member function to allow different allocator argument for the new string to be used or the obtained string copy. Add an overload of the `str(s)` member function that moves from a string and add `pilfer()` and `view()` member function:

```
// ??, members:
basic_stringbuf<charT, traits, Allocator>* rdbuf() const;

template<class SAlloc=Allocator>
basic_string<charT,traits,AllocatorSAlloc> str(const SAlloc& sa=SAlloc()) const;
template<class SAlloc=Allocator>
void str(const basic_string<charT, traits, SAllocator>& s);

void str(basic_string<charT, traits, Allocator>&& s);
basic_string<charT,traits,Allocator> pilfer();
basic_string_view<charT, traits> view() const;
```

6.2.1 27.8.3.1 basic_istream constructors [istream.cons]

Change the constructor specification to allow a string copy with a different allocator.

```
template<class SAlloc=Allocator>
explicit basic_istream(
    const basic_string<charT, traits, SAllocator>& str,
    ios_base::openmode which = ios_base::in);
```

- 1 *Effects:* Constructs an object of class `basic_istream<charT, traits>`, initializing the base class with `basic_istream(&sb)` and initializing `sb` with `basic_stringbuf<charT, traits, Allocator>(str, which | ios_base::in)` (??).

Add the following constructor specification:

```
explicit basic_istream(
    const basic_string<charT, traits, Allocator>&& str,
    ios_base::openmode which = ios_base::in);
```

- 2 *Effects:* Constructs an object of class `basic_istream<charT, traits>`, initializing the base class with `basic_istream(&sb)` and initializing `sb` with `basic_stringbuf<charT, traits, Allocator>(std::move(str), which | ios_base::in)` (27.8.2.1).

6.2.2 27.8.3.3 Member functions [istream.members]

Add the allocator parameter to the following `str()` overloads:

```
template<class SAlloc=Allocator>
basic_string<charT,traits,SAllocator> str(const SAlloc& sa=SAlloc()) const;
```

- 1 *Returns:* `rdbuf()->str(sa)`.

```
template<class SAlloc=Allocator>
void str(const basic_string<charT, traits, SAllocator>& s);
```

2 *Effects:* Calls `rdbuf()->str(s)`.

Add the following specifications:

```
void str(basic_string<charT, traits, Allocator>&& s);
```

3 *Effects:* `rdbuf()->str(std::move(s))`.

```
basic_string<charT, traits, Allocator> pilfer();
```

4 *Returns:* `rdbuf()->pilfer()`.

```
basic_string_view<charT, traits> view() const;
```

5 *Returns:* `rdbuf()->view()`.

6.3 27.8.4 Adjust synopsis of `basic_ostringstream` [`ostringstream`]

Add a new constructor overload and change the one taking the string by copy to allow a different allocator for the copied from string:

```
// ??, constructors:
explicit basic_ostringstream(
    ios_base::openmode which = ios_base::out);
template<class SAlloc=Allocator>
explicit basic_ostringstream(
    const basic_string<charT, traits, SAllocator>& str,
    ios_base::openmode which = ios_base::out);

explicit basic_ostringstream(
    basic_string<charT, traits, Allocator>&& str,
    ios_base::openmode which = ios_base::out);

basic_ostringstream(const basic_ostringstream& rhs) = delete;
basic_ostringstream(basic_ostringstream&& rhs);
```

Change the `str()` member function to allow different allocator argument for the new string to be used or the obtained string copy. Add an overload of the `str(s)` member function that moves from a string and add `pilfer()` and `view()` member function:

```
// ??, members:
basic_stringbuf<charT, traits, Allocator>* rdbuf() const;

template<class SAlloc=Allocator>
basic_string<charT, traits, AllocatorSAlloc> str(const SAlloc& sa=SAlloc()) const;
template<class SAlloc=Allocator>
void str(const basic_string<charT, traits, SAllocator>& s);
```

```
void str(basic_string<charT, traits, Allocator>&& s);
basic_string<charT, traits, Allocator> pilfer();
basic_string_view<charT, traits> view() const;
```

6.3.1 27.8.4.1 basic_ostringstream constructors [ostringstream.cons]

Change the constructor specification to allow a string copy with a different allocator.

```
template<class SAlloc=Allocator>
explicit basic_ostringstream(
    const basic_string<charT, traits, SAllocator>& str,
    ios_base::openmode which = ios_base::out);
```

- 1 *Effects:* Constructs an object of class `basic_ostringstream<charT, traits>`, initializing the base class with `basic_ostream(&sb)` and initializing `sb` with `basic_stringbuf<charT, traits, Allocator>(str, which | ios_base::out)` (??).

Add the following constructor specification:

```
explicit basic_ostringstream(
    const basic_string<charT, traits, Allocator>&& str,
    ios_base::openmode which = ios_base::out);
```

- 2 *Effects:* Constructs an object of class `basic_ostringstream<charT, traits>`, initializing the base class with `basic_ostream(&sb)` and initializing `sb` with `basic_stringbuf<charT, traits, Allocator>(std::move(str), which | ios_base::out)` (27.8.2.1).

6.3.2 27.8.4.3 Member functions [ostringstream.members]

Add the allocator parameter to the following `str()` overloads:

```
template<class SAlloc=Allocator>
basic_string<charT, traits, SAllocator> str(const SAlloc& sa=SAlloc()) const;
```

- 1 *Returns:* `rdbuf()->str(sa)`.

```
template<class SAlloc=Allocator>
void str(const basic_string<charT, traits, SAllocator>& s);
```

- 2 *Effects:* Calls `rdbuf()->str(s)`.

Add the following specifications:

```
void str(basic_string<charT, traits, Allocator>&& s);
```

3 *Effects:* `rdbuf()->str(std::move(s)).`

```
basic_string<charT,traits,Allocator> pilfer();
```

4 *Returns:* `rdbuf()->pilfer().`

```
basic_string_view<charT, traits> view() const;
```

5 *Returns:* `rdbuf()->view().`

6.4 27.8.5 Adjust synopsis of `basic_stringstream` [`stringstream`]

Add a new constructor overload and change the one taking the string by copy to allow a different allocator for the copied from string:

```
// ??, constructors:
explicit basic_stringstream(
    ios_base::openmode which = ios_base::out | ios_base::in);
template<class SAlloc=Allocator>
explicit basic_stringstream(
    const basic_string<charT, traits, SAllocator>& str,
    ios_base::openmode which = ios_base::out | ios_base::in);

explicit basic_stringstream(
    basic_string<charT, traits, Allocator>&& str,
    ios_base::openmode which = ios_base::in | ios_base::out);

basic_stringstream(const basic_stringstream& rhs) = delete;
basic_stringstream(basic_stringstream&& rhs);
```

Change the `str()` member function to allow different allocator argument for the new string to be used or the obtained string copy. Add an overload of the `str(s)` member function that moves from a string and add `pilfer()` and `view()` member function:

```
// ??, members:
basic_stringbuf<charT, traits, Allocator>* rdbuf() const;

template<class SAlloc=Allocator>
basic_string<charT,traits,AllocatorSAlloc> str(const SAlloc& sa=SAlloc()) const;
template<class SAlloc=Allocator>
void str(const basic_string<charT, traits, SAllocator>& s);

void str(basic_string<charT, traits, Allocator>&& s);
basic_string<charT,traits,Allocator> pilfer();
basic_string_view<charT, traits> view() const;
```

6.4.1 27.8.4.1 basic_stringstream constructors [stringstream.cons]

Change the constructor specification to allow a string copy with a different allocator.

```
template<class SAlloc=Allocator>
explicit basic_stringstream(
    const basic_string<charT, traits, SAllocator>& str,
    ios_base::openmode which = ios_base::out | ios_base::in);
```

- 1 *Effects:* Constructs an object of class `basic_stringstream<charT, traits>`, initializing the base class with `basic_istream(&sb)` and initializing `sb` with `basic_stringbuf<charT, traits, Allocator>(str, which)`.

Add the following constructor specification:

```
explicit basic_stringstream(
    const basic_string<charT, traits, Allocator>&& str,
    ios_base::openmode which = ios_base::in | ios_base::out);
```

- 2 *Effects:* Constructs an object of class `basic_stringstream<charT, traits>`, initializing the base class with `basic_stream(&sb)` and initializing `sb` with `basic_stringbuf<charT, traits, Allocator>(std::move(str), which)` (27.8.2.1).

6.4.2 27.8.4.3 Member functions [stringstream.members]

Add the allocator parameter to the following `str()` overloads:

```
template<class SAlloc=Allocator>
basic_string<charT, traits, SAllocator> str(const SAlloc& sa=SAlloc()) const;
```

- 1 *Returns:* `rdbuf()->str(sa)`.

```
template<class SAlloc=Allocator>
void str(const basic_string<charT, traits, SAllocator>& s);
```

- 2 *Effects:* Calls `rdbuf()->str(s)`.

Add the following specifications:

```
void str(basic_string<charT, traits, Allocator>&& s);
```

- 3 *Effects:* `rdbuf()->str(std::move(s))`.

```
basic_string<charT, traits, Allocator> pilfer();
```

- 4 *Returns:* `rdbuf()->pilfer()`.

```
basic_string_view<charT, traits> view() const;
```

5 *Returns:* `rdbuf()->view()`.

7 Appendix: Example Implementations

The given specification has been implemented within a recent version of the `sstream` header of gcc6. Modified version of the headers and some tests are available at

https://github.com/PeterSommerlad/SC22WG21_Papers/tree/master/workspace/Test-basic_stringbuf_efficient/src.